

ROTA AB, Spain

WMO: 084490

Lat: 36.639N

Lon: 6.332W

Elev: 84

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 13025

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 37.5	(c) 39.9	(d) 26.5	(e) 20.5	(f) 45.9	(g) 30.3	(h) 24.5	(i) 46.7	(j) 28.9	(k) 55.6	(l) 25.1	(m) 55.5	(n) 7.5	(o) 40	(p) 0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.6	(c) 92.8	(d) 69.9	(e) 89.6	(f) 69.2	(g) 86.3	(h) 68.8	(i) 74.5	(j) 83.7	(k) 73.0	(l) 81.7	(m) 71.9	(n) 80.5	(o) 13.8	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.7	117.4	78.4	70.1	110.9	77.0	69.4	108.3	76.5	37.9	83.9	36.7	81.9	35.7	81.0	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.7	19.3	33.6	98.6	3.3	3.0	31.3	100.8	29.4	102.6	27.5	104.3	25.1	106.5		
			30.8	76.6	3.3	3.1	28.4	78.8	26.5	80.6	24.6	82.3	22.2	84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	52.5	53.8	57.8	61.0	66.3	71.9	75.2	76.2	72.3	67.7	59.3	54.4
	DBStd	9.44	4.53	4.56	4.18	4.22	4.89	4.68	4.38	4.47	4.24	4.39	5.11	4.78
	HDD50	59	25	16	2	0	0	0	0	0	0	0	2	14
	HDD65	1635	388	314	224	133	41	1	0	0	1	23	182	328
	CDD50	5205	103	122	245	331	505	658	780	813	669	549	279	151
	CDD65	1304	0	0	1	14	81	209	315	348	219	107	10	0
	CDH74	9041	0	1	18	111	546	1449	2413	2770	1268	450	15	0
	CDH80	2955	0	0	1	14	142	504	855	996	364	79	0	0
Wind	WSAvg	8.9	8.3	9.1	9.9	10.1	9.8	9.6	8.6	8.4	8.3	8.3	8.3	8.5
Precipitation	PrecAvg	21.1	2.8	2.4	2.2	1.7	1.1	0.3	0.0	0.1	0.8	3.0	3.7	3.1
	PrecMax	35.7	9.1	10.8	10.2	4.8	3.9	2.1	0.1	1.3	3.3	9.6	14.2	13.4
	PrecMin	10.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	6.6	2.1	2.0	2.0	1.2	1.1	0.5	0.0	0.2	0.9	2.6	3.1	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.5	70.2	75.6	82.2	88.2	94.6	95.4	96.6	91.5	86.0	75.7	68.4
		MCWB	58.3	57.9	60.3	62.9	65.0	68.5	70.1	70.1	69.9	67.1	63.3	60.2
	2%	DB	64.3	66.3	72.0	77.2	84.1	89.5	91.7	92.8	88.0	82.2	72.2	66.2
		MCWB	58.4	56.8	59.3	60.6	64.0	66.9	69.9	71.2	69.1	66.1	63.0	60.0
	5%	DB	62.5	64.3	69.6	73.3	80.2	85.9	88.2	89.3	84.3	78.8	70.0	64.5
		MCWB	57.5	56.6	58.6	60.1	62.8	66.4	69.3	70.9	68.5	65.6	62.5	59.4
	10%	DB	60.8	62.3	66.2	69.7	76.3	82.3	84.4	85.7	80.7	75.5	67.9	62.7
		MCWB	55.9	56.5	57.3	59.7	62.6	66.3	68.7	70.5	68.2	65.3	61.8	58.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	61.5	63.9	66.3	69.6	72.8	75.5	77.3	74.6	71.2	68.2	63.9
		MCDB	64.0	64.9	72.0	74.7	80.0	84.4	86.4	86.3	82.7	75.7	71.1	66.0
	2%	WB	60.1	59.8	61.5	63.8	67.3	70.7	73.7	75.0	72.7	69.5	65.7	62.0
		MCDB	62.5	63.2	67.7	70.9	76.9	80.3	83.1	84.1	80.4	75.0	69.6	64.1
	5%	WB	58.4	58.6	60.1	62.1	65.6	69.4	72.4	73.6	71.2	68.4	64.1	60.5
		MCDB	61.2	62.1	65.6	68.8	74.2	78.6	81.4	82.5	78.7	74.3	68.2	63.3
	10%	WB	56.5	56.9	58.9	60.8	64.4	68.3	71.2	72.3	70.0	67.2	62.5	59.0
		MCDB	59.9	61.0	64.1	67.1	72.3	77.6	80.0	81.1	76.8	73.3	66.6	62.4
Mean Daily Temperature Range		MDBR	14.9	15.2	15.3	14.6	15.0	15.1	15.5	15.6	14.7	14.9	14.4	14.3
	5% DB	MCDBR	16.0	18.3	21.5	22.5	22.9	23.1	22.7	21.9	21.3	20.3	16.5	15.1
		MCWBR	11.5	11.9	12.8	12.2	11.3	10.3	10.3	10.0	10.0	10.3	10.3	10.4
	5% WB	MCDBR	12.1	14.2	16.6	16.6	17.5	17.3	17.4	17.7	15.2	14.9	12.7	11.2
MCWBR		10.1	10.7	11.4	10.9	10.2	9.1	9.3	9.5	8.6	8.8	9.3	9.4	
Clear-Sky Solar Irradiance	taub	0.332	0.353	0.392	0.388	0.401	0.414	0.401	0.419	0.406	0.383	0.351	0.337	
	taud	2.503	2.447	2.326	2.362	2.353	2.333	2.360	2.320	2.379	2.445	2.498	2.504	
	Ebn at Noon	273	279	277	284	280	275	278	271	269	265	263	263	
	Edh at Noon	26	31	38	38	39	40	39	40	36	30	26	24	
All-Sky Solar Radiation	RadAvg	857	1142	1524	1916	2271	2492	2529	2233	1756	1271	918	755	
	RadStd	84	124	138	115	119	100	45	68	85	101	73	70	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112

Nomenclature: See separate page